SDUHSD CAREER TECHNICAL EDUCATION

Introduction to Engineering Techniques

Campus: Canyon Crest Academy

CTE Industry Sector: Engineering & Architecture	**CTE Pathway:** Engineering Technology	**Course Sequence: Course 1 - Introductory**
10 credits	**UC-G approved elective**	**Prerequisites:** none

Recommended Pathway Sequence:

Course 1 - Introductory	Course 2 - Concentrator	Course 3- Capstone	Enrichment
Introduction to Engineering Techniques	Digital Electronics	Advanced Engineering Technology	Robotics and Engineering Technology

Course Description:
Introduction to Engineering Techniques provides students with tools to pursue further study in engineering. This includes but is not limited to the study of strengths of materials, electronic circuits, mathematical analysis, and engineering design. Coursework is heavily project based learning as well as reading texts and writing. Assignments will be from various fields of engineering including Mechanical, Civil, Electrical, and Aerospace. The class is intended for those interested in exploring engineering as a career or seeking a CTE class that is connected to STEM. This course is UC Approved.
*This course was previously called Principles of Engineering.

- Standards for Engineering and Architecture
- http://www.corestandards.org/ELA-Literacy/RST/11-12/
- http://www.nextgenscience.org

Course Syllabi:
CCA

Grading:
A variety of assessments will be used including journal entries, projects and self and peer assessments. For example students will be assessed on their journal entries which include warm-up exercises to calculate the force on a bridge truss. Another example assessment includes student

demonstration of a project implementing a mechanical, electrical, and control mechanism to sort different colored marbles. Another example assessment includes a student self-evaluation of the design of his or her material strength project. Another example assessment includes peer review and voting of students' simple machine energy transfer.

Workload: (High, Medium, Low)
Class setting: This relates directly to how much the student is expected to work independently or in a group requiring self-initiated work, problem solving, etc versus and highly guided and structured class).
(Highly independent, Moderately Independent, Structured). Hands-on projects accompany each unit.
*Each year all CTE courses are reviewed by industry professionals within the pathway to ensure the most current, relevant experience for our students

Textbooks or Supplemental Materials for this course: Supplemental Materials provided electronically.

Software and equipment knowledge specific to this course: Autodesk Inventor, Autodesk Revit, LoggerPro, MDsolids, Scilab, Westpoint Bridge Designer, BrickxC Integrated Development Environment

Units of Study, Large Projects, Work-based Learning/Industry Connections.
Every February - students showcase projects with local businesses and community members

1. History of Engineering, Engineering Careers, Intellectual Property (Engineering Poster project)
2. Engineering Design Process and Modeling (Product Teardown and modeling project - lawn sprinkler)
3. Simple Machines (Simple Machine Energy Transfer project)
4. Kinematics (Projectile Motion Device project)
5. Statics and Structures (Bridge design project)
6. Architecture (Coffee Shop design project)
7. Control Systems (Marble Sorter project)

Framework Concept	Engineering Technology	Leadership opportunities 1 Standards for Career Ready Practice (1 or 2) 2 Student Clubs 3 Portfolio building & Presentation
History of Engineering and Engineering Careers -	• B1.0: Communicate and interpret information clearly in industry-standard visual and written formats	1. Apply appropriate technical skills and academic knowledge. 4. Apply technology to enhance productivity.

A sentence or two about what the project is. **Engineering Poster Project**		5. Utilize critical thinking to make sense of problems and persevere in solving them. 10. Demonstrate creativity and innovation.
Engineering Design Process and Modeling **Product Teardown and modeling project - lawn sprinkler**	● B2.0: Demonstrate the sketching process used in concept development ● B6.0: Employ the design process to solve analysis and design problems	
Simple Machines **Simple Machine Energy Transfer project**	● B5.0: Understand how the principles of force, work, rate, power, energy, and resistance relate to mechanical, electrical, fluid, and thermal engineering systems ● B6.0: Employ the design process to solve analysis and design problems ● B7.0: Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance	
Kinematics **Projectile Motion Device project**	● B4.0: Understand the concepts of physics that are fundamental to engineering technology ● B6.0: Employ the design process to solve analysis and design problems ● B7.0: Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance	
Statics and Structures **Bridge design project**	● B9.0: Understand the fundamentals of systems and market influences on products as they are developed and released to production ● B6.0: Employ the design process to solve analysis and design problems ● B7.0: Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance	
Architecture **Coffee Shop design project**	● B6.0: Employ the design process to solve analysis and design problems ● B7.0: Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance	

	● B11.0 Understand the methods of creating both written and digital portfolios.	
Control Systems **Marble Sorter project**	● B8.0 Understand fundamental control system design and develop systems that complete pre-programmed tasks ● B6.0: Employ the design process to solve analysis and design problems ● B7.0: Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance	